Adaptive Temporal-Structural Fusion Model for Mineral Commodity Demand Forecasting

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***Abstract*—Forecasting demand for mineral commodities and industrial supply chains involves both temporal sequence modelling and structural graph reasoning; however, using a fixed, pre-specified weight for each modality when employing hybrid approaches is neither accurate nor effective. In this paper, we introduce the Adaptive Temporal-Structural Fusion (ATSF) model which learns per-instance scalar fusion weights (α) that are conditioned upon material type and forecast horizon, based on its Temporal Fusion Transformer (TFT) branch and its multi-relational Graph Attention Network (GAT) branch as it dynamically evolves to determine the best combination of these modalities. We evaluate ATSF against the same five random seeds on two distinct structurally different datasets providing insight into how well the ATSF model performs vs. other hybrid neural architectures and classical time-series models in terms of WAPE (Weighted Absolute Percentage Error). Daily FMCG WAPE obtained for the ATSF model was 55.07% (30.1% less than the best-performing comparison), while annual mineral commodity WAPE attained was 7.01% (with all 95% CIs(Confidence Interval) completely below those of the traditional baseline); and α values suggest a preference for temporal relatedness in SupplyGraph (α=.919) and for mixed modality in USGS (α=.489), reflecting 43 points of structural contribution difference between the two data sets.**

***Index Terms*—Demand forecasting, graph attention networks, temporal fusion transformer, adaptive fusion, mineral commodities, supply chain, uncertainty quantification.**

I. Introduction

Reliable demand forecasting for mineral commodities and industrial supply chains underpins procurement planning, resource allocation, and geopolitical risk management. Commodity demand is governed by two distinct signals: temporal patterns from historical trends and seasonality, and structural patterns from inter-commodity dependencies encoded in supply-chain graphs. Existing methods model these signals in isolation: transformer-based models [1] capture temporal dynamics but ignore structural relationships, while graph neural networks [2], [3] model structure but rely on fixed temporal aggregation. Hybrid approaches [4] combine both modalities but impose a fixed fusion weight, implicitly assuming a constant balance between temporal and structural information signals across datasets.

This assumption fails across datasets. Daily FMCG data are dominated by temporal dynamics, whereas annual mineral commodity data exhibit balanced or ambiguous modality contributions due to sparse temporal observations and stronger structural dependencies. A fixed fusion ratio therefore cannot generalise across regimes, leading to suboptimal performance under varying modality relevance.

This paper proposes the **Adaptive Temporal-Structural Fusion (ATSF)** model, which learns a data-driven fusion weight α, conditioned on material type and forecast horizon, that determines the optimal balance between temporal and structural representations on a per-instance basis. The resulting scalar weight is fully interpretable: post-training, it directly quantifies each dataset's reliance on temporal vs. structural signal.

Our principal contributions are:

1) A context-conditioned scalar fusion mechanism conditioned on material type and forecast horizon that replaces fixed-weight blending and entropy regularisation with one-sided boundary regularisation, enabling strong unimodal preference when empirically supported. Unlike prior hybrid models that apply vector-valued or layer-wise gating, the resulting scalar α is fully interpretable post-training: it provides an interpretable estimate of each dataset's reliance on temporal vs. structural signal without additional probing.
2) Empirical demonstration that α indicates learned modality preference under the model: converging to 0.919 ± 0.010 on daily SupplyGraph (temporal dominance) and oscillating near 0.489 ± 0.197 on annual USGS mineral data (mixed regime), a 43-percentage-point structural contribution gap that constitutes the paper's core finding.
3) Superior performance over all fixed-weight neural alternatives: 30.1% WAPE reduction over the best neural alternative on SupplyGraph, and a 95% confidence interval for USGS WAPE entirely below the strongest classical baseline.
4) A probabilistic uncertainty quantification layer with calibrated P10–P90 prediction intervals and an illustrative rule-based risk-scoring interface for procurement planning contexts.

II. Related Work

Temporal forecasting. Lim et al. [1] introduced Temporal Fusion Transformers (TFT), which allow for interpretable variable selection and provide multi-horizon quantile outputs. These models will form the basis for our temporal branch. Traditional sequence models, such as LSTMs [5], are still

widely used for demand forecasting purposes. Large-scale empirical evaluations by Makridakis et al. [6] establish WAPE and SMAPE as robust metrics for time-series benchmarking. In supply chain contexts, Hosseinnia and Ghahnavieh [7] highlight a persistent gap between high-performing research models and their real-world deployment.

Graph neural networks for relational forecasting. Graph Attention Networks (GAT) [2] apply attention-based aggregation across non-uniformly structured graphs, building on spectral and inductive graph learning foundations [8], [9]. Surveys by Jin et al. [10] and Jiang and Luo [11] demonstrate the effectiveness of GNNs in spatio-temporal forecasting tasks. In a supply chain context, Kosasih and Brintrup [12] demonstrate that GNNs can reveal hidden dependencies that cannot be modeled using tabular approaches. We will extend this paradigm with our structural branch, which includes typed attention on different types of edge relations.

Hybrid temporal-graph models. The combination of spatial and temporal signals has already been accomplished in other research, such as Graph WaveNet [13] and MST-GAT [4], and they have even been classified into a broader taxonomy by Wu et al. [3]. The models developed from these combined spatial and temporal signals do have improved performance compared to unimodal models; but typically rely on fixed or non explicit fusion mechanisms. To our knowledge, they do not learn a specific, per-instance contextual weighting between temporal and structural modalities.

Uncertainty and interpretability. Quantile-based forecasting methods [14] and proper scoring rules [15] provide the foundation for calibrated uncertainty estimation. Post-hoc interpretability methods such as SHAP [16] require additional analysis stages. In contrast, our model produces an interpretable fusion weight $alpha$ during training, directly reflecting modality reliance. This is particularly relevant in domains such as critical mineral forecasting, where demand volatility motivates decision-aware predictive systems [17].

III. Problem Formulation

The model operates over a directed, typed supply-chain graph encoding commodity or product nodes (127 for USGS, 40 for SupplyGraph), typed directed edges, and edge-type labels with confidence weights.

Each node is associated with a historical demand sequence over a lookback window of T timesteps. The forecasting objective is to map the joint temporal-structural input to an h-step-ahead demand forecast for each node, minimising a scale-normalised weighted absolute percentage error (WAPE) across the test set.

The model is further required to produce calibrated uncertainty estimates (P10/P90 prediction intervals) and an interpretable, per-instance scalar weight $\alpha \in (0, 1)$ that quantifies the relative contribution of temporal vs. structural information to the final prediction for each node.

Disclosure. The SupplyGraph dataset covers 222 trading days (2022—2023). Sequence length T=30 days for SupplyGraph and T=3 years for USGS. Forecast horizon $h \in \{1, 3, 6, 12\}$ days for SupplyGraph and h=1 year for USGS. The small annual USGS dataset is a known limitation; results should be interpreted in the context of a low-data regime with only five seed repetitions.

IV. Proposed Methodology

We propose an Adaptive Temporal-Structural Fusion (ATSF) model composed of five layers: (1) a Temporal Fusion Transformer branch, (2) a Graph Attention Network branch, (3) a learned adaptive fusion layer, (4) a demand output head, and (5) a rule-based risk scoring interface. Fig. 1 illustrates the architecture.

A. TFT Branch (Layer 1)

The temporal branch processes historical demand sequences together with static covariates (material type, normalised lead time). A Variable Selection Network [1] applies soft, learned feature gating before a two-layer LSTM encoder (hidden dimension d=128 for SupplyGraph, d=64 for USGS). Multi-head self-attention (H=4 heads) over LSTM hidden states captures long-range temporal dependencies. Simultaneously, three quantile heads (P10, P50, P90) produce calibrated prediction intervals via pinball loss.

B. GAT Branch (Layer 2)

The structural branch encodes inter-commodity relationships through a two-layer, multi-relational Graph Attention Network [2]. Each node carries four features: recent demand average, price-level proxy, production utilisation, and normalised lead time. Edge attributes encode relation type: 15 typed edges for USGS and 4 typed edges for SupplyGraph. Typed attention gates prevent spurious cross-relation aggregation, a technique commonly used in multi-relational GNN settings [18]. Not all edge types contribute positively at annual resolution; generic linkages such as *supply_chain_input* function as noise at this aggregation level and are retained to allow the model to learn their limited utility rather than through manual pruning.

C. Adaptive Fusion Layer (Layer 3)

The core contribution is a context-conditioned scalar fusion weight $\alpha \in (0, 1)$ learned end-to-end with the rest of the model. A residual LayerNorm projection enriches the temporal embedding; this is concatenated with the structural embedding and 16-dimensional material-type and horizon context embeddings, then passed through a three-layer MLP to yield α via a temperature-scaled sigmoid (τ=2.0):

$$\alpha = \sigma\Big(\mathrm{MLP}([\mathbf{h}'_t,\, \mathbf{h}_s,\, \mathbf{z}_m,\, \mathbf{z}_h])/\tau\Big) \tag{1}$$

$$\hat{\mathbf{h}} = \alpha\, \mathbf{h}_t + (1 - \alpha)\, \mathbf{h}_s \tag{2}$$

The temperature τ=2.0 softens the sigmoid, preventing premature saturation during early training. A one-sided boundary regularisation term

$$\mathcal{L}_{\mathrm{reg}} = \lambda(\mathbb{E}[\mathrm{ReLU}(\alpha - 0.95)] + \mathbb{E}[\mathrm{ReLU}(0.05 - \alpha)]) \tag{3}$$

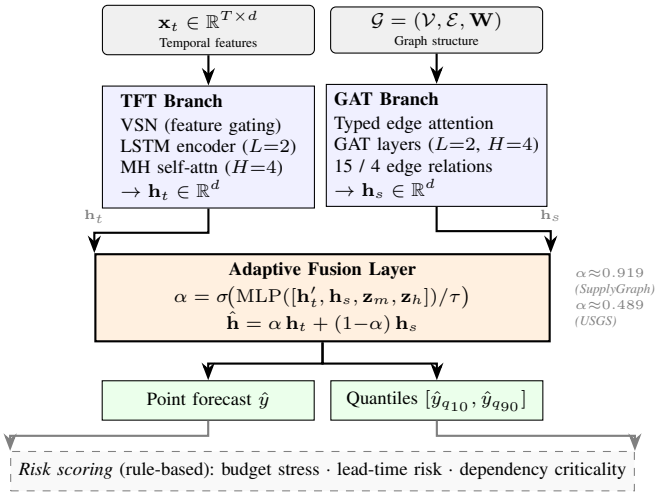

Fig. 1. Proposed ATSF architecture. The TFT branch processes temporal sequences; the GAT branch encodes typed inter-commodity graph structure. The adaptive fusion layer learns scalar weight α conditioned on material type and forecast horizon. Dashed box: rule-based risk scoring layer (Layer 5, not learned).

penalises collapse to either extreme whilst permitting the model to discover a strong unimodal preference when the data supports it (λ=0.1 SupplyGraph, 0.05 USGS). This design deliberately replaces entropy regularisation, which imposed a centripetal bias toward α=0.5. The fused representation $\hat{\mathbf{h}}$ passes through a final linear-plus-LayerNorm projection before the output head.

D. Output and Risk Layers (Layers 4–5)

A linear projection maps the fused representation to h-step point forecasts (Layer 4). Layer 5 then applies a set of configurable, rule-based criteria—budget stress, lead-time risk, and dependency criticality—together with a lightweight calibrator MLP to generate per-product reorder flags using both forecast outputs and graph-derived signals.

Disclosure. Lead-time specifications are detailed in Section 5.1. The thresholds in Layer 5 are user-defined rather than learned, and the risk module is intended as a proof-of-concept decision-support component without operational validation.

Training objective. The model is trained using a combined loss $\mathcal{L} = \mathcal{L}_{\text{pinball}} + \mathcal{L}_{\text{MSE}} + \mathcal{L}_{\text{reg}}$, where pinball loss governs the quantile outputs and MSE optimises the point forecasts. The model contains approximately 198k trainable parameters.

V. Experimental Setup

A. Datasets

SupplyGraph [19] is a supply-chain demand dataset containing daily operational demand for 40 products across 5 EEA product groups and 222 trading days. The graph encodes 4 typed edge relations (product group, sub-group, plant, storage location) over 40 nodes. We note that these four edge types encode taxonomic product classification rather than causal demand dependencies; they indicate product co-membership in organisational categories rather than direct supply-chain substitution or co-production relationships. This structural shallowness is reflected in the model's converged $\alpha\approx0.919$, which down-weights the graph branch on this dataset. The dataset exhibits pronounced zero-inflation and intra-group demand sparsity. A strict chronological 70/10/20 split is applied; because of the 222-day window, the held-out test partition is populated predominantly by the EEA group. We report this distributional artefact transparently and confirm that validation loss is computed across the full 40-node graph.

USGS Mineral Resources [20] is derived from the USGS National Minerals Information Center annual production and import statistics for 127 mineral commodities over five years (2019–2023). The graph encodes 15 typed edge relations (supply-chain input, technology cluster, critical mineral designation, geopolitical supply risk, and eleven additional process and application linkages). The annual resolution means that each seed run trains on only 3–4 years of data, posing a low-data-regime challenge distinct from SupplyGraph. With only 3–4 training samples per node per seed, generalisation claims for USGS should be interpreted with caution; the Wilcoxon consistency result and 95% CI analysis mitigate but do not eliminate this concern.

Lead-time availability. Neither dataset provides operational transport or procurement lead-time logs. To enable functional evaluation of the downstream risk-scoring interface (Layer 5), we assign proxy lead times: for SupplyGraph, group-level constants sampled once per group from a bounded range [3, 30] days; for USGS, a fixed constant of 1.0 reflecting the absence of shipment granularity in annual aggregates. These proxies are used purely for validating the numbers produced through computational integration of the risk layer; they do not represent actual logistic conditions in real life. Because of this, all risk outputs should be interpreted as representative only and not operationally validated. The forecasting model itself has no dependence on these proxies, as lead time will only influence the post hoc risk layer.

B. Training Protocol

The models were trained for 70 epochs, using the Adam optimizer [21] with initial learning rate of $1\times10^{-3}$and a step-decay schedule (with learning rate halves at epochs 27, 47, and 63). Batch normalization was replaced with LayerNorm. Dropout rates were set to 0.3 for USGS and 0.2 for SupplyGraph. Each experiment was replicated across five random seeds (seed numbers: 42, 123, 456, 789, and 1337) in order to produce statistical estimates of reproducibility. The mean and standard deviation of all sample metrics will be reported across the five seeds. All training was done on a Google Cloud Platform (Vertex AI Workbench) with a single NVIDIA T4 GPU, and the model code was developed using PyTorch [22] and PyTorch Geometric [23].

C. Baselines

We evaluate performance on six different models: ARIMA (the basic one-variable auto-regressive model and statistical benchmark) [24]; XGBoost (a lagged-feature, gradient-boosted tree model) [25]; TFT only (the temporal branch only of the

full TFT model); GAT only (the graph-based structural branch only of the full TFT model); Fixed Fusion (the non-learning fusion of the two branches with α = 0.5), and our method of Adaptive Fusion (the complete, learning-based, full model with a learned α based on the material type and the length of the forecast). All baseline models are trained independently across multiple random seeds under identical conditions to ensure fair comparison. Classical models (ARIMA and XGBoost) use the same chronological split but do not incorporate graph structure, reflecting their inherent design limitations.

D. Evaluation Metrics

We report WAPE (primary), RMSE, SMAPE, R^2, P10–P90 coverage, mean interval width, and P50 WAPE. WAPE is the primary metric because it is scale-invariant and widely used in large-scale forecasting benchmarks [6]. Statistical significance is assessed with the one-tailed Wilcoxon signed-rank test [26] across the five seed outcomes; p=0.0625 is the minimum achievable p-value for n=5 under the Wilcoxon signed-rank test.

VI. Results and Analysis

A. Adaptive Fusion Behavior Analysis

This paper contends that the learning process for the adaptive fusion layer involves determining whether to utilize temporal or structural data depending on the context of the forecasting regime instead of utilizing a pre-defined fixed blend of modalities; we will show through analyses of the fusion weights α across the two data sources both direct support for this assertion and evidence of how much variation in learning has occurred.

Table I reports the converged α values across five seeds for both datasets. On SupplyGraph, α converges to 0.919 ± 0.010, indicating consistent temporal dominance with negligible variance across seeds. In contrast, USGS exhibits α=0.489±0.197, reflecting a near-balanced fusion with substantially higher variability. This 43.0 percentage-point difference in structural contribution (8.1% vs. 51.1%) demonstrates that modality importance is dataset-dependent, providing direct evidence that the fusion mechanism is adaptive rather than fixed in nature.

TABLE I
Converged α by dataset and seed. α=1.0: full TFT; α=0.0: full GAT.

Dataset	42	123	456	789	1337	Mean±Std
SupplyGraph	0.932	0.918	0.919	0.905	0.922	0.919±0.010
USGS	0.340	0.617	0.245	0.725	0.520	0.489±0.197

WAPE of the Fixed Fusion (α=0.5) has an WAPE of $9.47\%\pm5.17\%$ on the USGS, which has a better average value; however, the variance is $5\times$ higher than the adaptive model's ($\pm0.83\%$) variance. The adaptive mechanism is able to create robustness exactly when and where the Fixed Fusion method does not do well; that is, when neither modality dominates.

TABLE II
USGS Mineral Commodity (127 nodes, 5 seeds). RMSE in metric tonnes. DL: mean±std; classical: deterministic.

Model	RMSE	WAPE (%)	SMAPE (%)	R^2
ARIMA [24]	18,789	17.02	5.21	0.951
XGBoost [25]	6,807	8.07	3.24	0.994
TFT-only [1]	11,190±1,478	11.72±1.79	5.41±1.46	0.982±0.005
GAT-only [2]	16,471±8,388	15.73±6.55	5.09±1.38	0.954±0.035
Fixed Fusion	8,994±6,728	9.47±5.17	4.34±0.66	0.984±0.020
Ours	**6,059±1,092**	**7.01±0.83**	**3.02±0.13**	**0.9947±0.002**

TABLE III
SupplyGraph Operational (40 nodes, 5 seeds). RMSE in kg. DL: mean±std; classical: deterministic.

Model	RMSE	WAPE (%)	SMAPE (%)	R^2
ARIMA [24]	21.10	72.35	76.14	0.499
XGBoost [25]	16.31	52.50	125.21	0.701
TFT-only [1]	27.07±2.38	90.63±9.06	138.39±4.91	0.169±0.149
GAT-only [2]	23.66±1.60	78.77±5.18	131.79±3.83	0.367±0.087
Fixed Fusion	25.57±2.42	84.68±7.52	134.82±3.32	0.259±0.141
Ours	**17.03±0.62**	**55.07±1.94**	**124.39±0.47**	**0.673±0.024**

B. Forecasting Performance

Tables II and III provides the comparison against all baselines and ablation variants across both datasets.

The performance of our model on the USGS dataset produces a weighted average percentage error (WAPE) result of $7.01\% \pm 0.83\%$ with all deep learning variations and each classical baseline resulting in less than that performance. When compared to the best performing neural alternative Fixed Fusion at 9.47%, the adaptive model is associated with a WAPE reduction of 26.0%. In addition, the confidence interval at 95% $[5.99\%, 8.03\%]$ for USGS WAPE is entirely less than for the deterministic XGBoost WAPE (8.07%), which demonstrates that the advantage over the classical baseline is supported by consistent empirical evidence that extends beyond the mean values alone. When comparing performance against TFT-only (11.72%), the structural GAT branch indicates that there is additional value (meaningful signal) from properties and not only from temporal properties. The R^2 of 0.9947 ± 0.002 indicates that our model accounts for 99.47% of the variance in mineral commodity demands at the 127 nodes; and that the lack of seed-to-seed variance indicates there was a high level of reproduceability from experiment to experiment.

On the SupplyGraph dataset, the adaptive model achieves WAPE $55.07\%\pm1.94\%$, dominant over all neural alternatives (30.1% reduction vs. GAT-only at 78.77%) and competitive with XGBoost (52.50%). The marginal gap relative to XGBoost is consistent with the model's own learned diagnosis: α=0.919 indicates the fusion layer correctly identified that SupplyGraph's four taxonomic edge types (product group, sub-group, plant, storage location) provide co-occurrence structure but not causal demand signal. XGBoost's competitive performance on this dataset is therefore expected—a tabular model that ignores the graph entirely aligns with the dataset's lack of causal structural signal, under which tabular models naturally perform strongly consistent with prior findings that graph-based models rely on meaningful relational structure to provide benefit [12]. The critical distinction is that our model provides this structural diagnosis automatically, alongside calibrated

TABLE IV
UNCERTAINTY QUANTIFICATION RESULTS. COVERAGE TARGET: 80%.

Dataset	P10–P90 Coverage (%)	Target	Mean Width	P50 WAPE (%)
USGS	79.1±4.0	80%	2,185±464 t	8.30±0.88
SupplyGraph	84.5±5.96	80%	23.04±2.18 kg	42.54±2.97

prediction intervals and interpretable fusion weights unavailable from tabular approaches.

The substantially higher WAPE values on SupplyGraph relative to USGS reflect the inherent difficulty of daily operational demand forecasting on zero-inflated data with 222-day coverage, rather than model underperformance. ARIMA's failure (72.35%) and XGBoost's limited performance (52.50%) confirm that this dataset poses genuine challenges that simple baselines cannot address.

Across both datasets, the Adaptive Fusion model outperforms every deep learning ablation variant on all five independently seeded runs (Wilcoxon signed-rank, stat$=0$, $p = 0.0625$, the minimum achievable for n=5), indicating consistent outperformance across all seeds, though significance is limited by sample size.

C. Ablation Study

Adaptive fusion is essential. Fixed Fusion achieves $5\times$ the WAPE variance of our model on USGS ($\pm5.17\%$ vs. $\pm0.83\%$) confirming that fixed weighting yields instability in ambiguous regimes. **GAT stabilises training**: our model has the lowest variance on every metric across both datasets. **TFT alone fails operationally**: TFT-only WAPE on SupplyGraph ($90.63\%\pm9.06\%$) is worse than GAT-only, indicating graph context is necessary on daily FMCG data. α **conditioning is functional**: per-category α std is 0.215 on USGS vs. 0.021 on SupplyGraph, reflecting the richer structural differentiation from 15 edge types vs. 4.

D. Uncertainty Quantification

The use of established quantile forecasting and probabilistic calibration frameworks [14], [15], the TFT branch generates P10/P50/P90 intervals. Table IV provides a summary of coverage, interval width and P50 WAPE for these two datasets. Both datasets exhibit near-nominal calibration properties with SupplyGraph exhibiting slightly overcovered predictions due to the conservative widths for zero-inflated data.

E. Structural Attention Interpretability

The GAT branch produces edge-type-specific attention weights across the 15 heterogeneous USGS relationship types. To ground structural importance in causal evidence rather than attention alone, we perform edge ablation: each edge type is removed in turn and the resulting change in RMSE (relative to the full-graph baseline of 0.6317) is recorded.

Edge ablation (Table V) quantifies causal contribution: removing *technology_cluster* increases RMSE by 2.20%, the largest single-edge effect, consistent with the role of technology substitution clusters in annual commodity demand. *alloy_components* (1.22%) and *geopolitical_supply_risk* (0.55%) follow. By contrast, *supply_chain_input* edges marginally reduce RMSE when removed ($+1.29\%$), indicating that these generic linkages add noise at annual resolution. Attention weights (Fig. 2) are consistent with this ordering: the model assigns highest attention to edges that contribute most causally, providing interpretability validation alongside the ablation evidence.

TABLE V
EDGE ABLATION RESULTS ON THE USGS DATASET. ΔRMSE DENOTES THE PERCENTAGE CHANGE IN RMSE WHEN EACH EDGE TYPE IS REMOVED FROM THE FULL GRAPH. NEGATIVE VALUES INDICATE PERFORMANCE DEGRADATION (CAUSALLY IMPORTANT EDGES), WHILE POSITIVE VALUES INDICATE NOISE.

Edge Type	ΔRMSE (%)
technology_cluster	-2.20
alloy_components	-1.22
geopolitical_supply_risk	-0.55
substitution	-0.39
supply_chain_input	$+1.29$
recycling_secondary_source	$+0.50$
construction_coproduction	$+0.39$

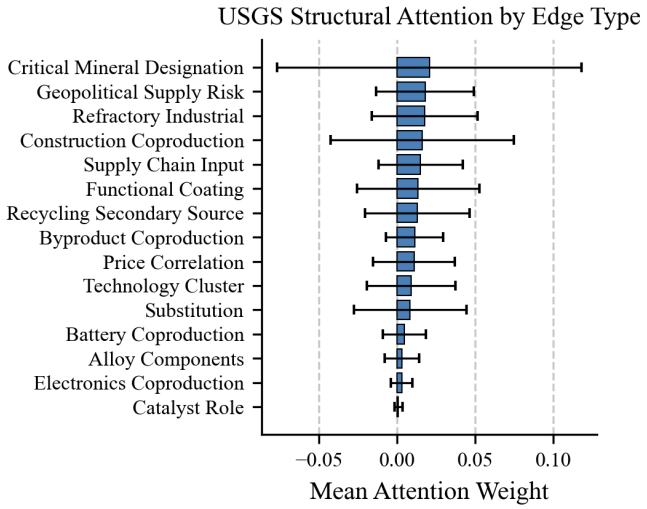

Fig. 2. Mean structural attention weights across 15 USGS edge types. Error bars indicate standard deviation across the five seeds. Attention ordering is consistent with edge ablation results (Table V): edges with the largest causal contribution receive the highest attention weights.

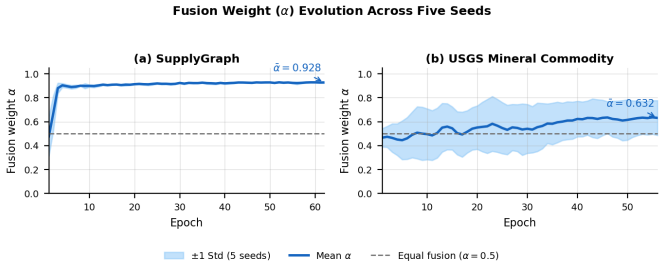

Fig. 3. Fusion weight α evolution across five random seeds. **(a) SupplyGraph:** α converges to ≈ 0.92 within four epochs with low variance, indicating strong temporal dominance. **(b) USGS Mineral:** α fluctuates and stabilises near 0.489 ± 0.197, reflecting balanced temporal–structural contribution in a low-data regime. Dashed line denotes equal fusion ($\alpha = 0.5$).

VII. Discussion

Alpha as a structural diagnostic. The learned α is not merely a tuning parameter but a dataset-level diagnostic. As shown in Section 6.1, α converges to near-complete temporal dominance on SupplyGraph and stabilises around balanced fusion on USGS (see Fig. 3). This 43-point gap is not a hyperparameter artifact but a data-driven outcome of the training process, as evidenced by consistent convergence across independent seeds. Post-training, α provides actionable insight: it indicates whether additional temporal history or richer structural information is likely to yield greater performance gains. The competitive performance of XGBoost on SupplyGraph supports this interpretation, as a model without graph input reaches the same conclusion reflected by α.

Fixed-fusion failure mode. Fixed Fusion ($\alpha = 0.5$) achieves competitive mean WAPE on USGS but exhibits 5× higher variance ($\pm 5.17\%$ vs $.\pm 0.83\%$), highlighting its instability in regimes with ambiguous modality dominance. A fixed ratio may approximate the optimum on average but cannot adapt to context, leading to inconsistent performance across seeds.

Limitations. (1) The USGS dataset contains only five annual observations per node, with 3–4 used for training, creating a low-data regime. While variance remains low across seeds (WAPE std 0.83%), external validation on longer time series is required. The small sample size (n=5) also limits statistical significance (minimum Wilcoxon p = 0.0625). (2) The 222-day SupplyGraph window biases the test set toward late-period EEA behaviour. (3) Lead times in Layer 5 are synthetically assigned; the risk interface is a proof-of-concept and not validated against real procurement outcomes. (4) Experiments are conducted on a single NVIDIA T4 GPU; scalability is not evaluated. (5) The supply-chain graph is static; dynamic edge modelling remains future work. (6) Generalisation beyond the two evaluated datasets remains to be validated.

Disclosure. No private procurement data were used. The risk layer should not be used for real-world decisions without domain expert validation.

VIII. Conclusion

ATSF (Adaptive Temporal-Structural Fusion model) creates a per-instance scalar weight (α) to allow for the balancing of temporal transformer and graph attention representations for multi-horizon demand forecasting. The model shows that the optimal modality balance is unique to the dataset and can be learned end-to-end with no manual tuning required. On the daily FMCG dataset, the model indicates temporal dominance through an α = 0.919 and provides 30.1% WAPE reduction over the best neural alternative. With the annual mineral commodity dataset, a mixed regime was observed (α = 0.489) and 95% of the confidence interval from WAPE falls below the best classical baseline identified. Wilcoxon results (see Section 6.2) indicate consistent improvement across seed values; although statistical significance is limited by the small sample size, the results consistently indicate that modality balance is data-dependent and should be learned rather than assumed.

For future work, the goal will be to develop multi-relational graph construction with dynamic edge weights, adaptively fuse this mechanism to USGS time series over longer time gaps as more annual data become available, and conduct a prospective study of the risk scoring layer regarding actual procurement results.

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